December 07, 2025 12:23 PM · Model: gpt-5-1 · 16 messages

USER

Give me a list of 40 of the lesser known sources of magic as a precursor to experimental science considered by Lynn Thorndike in his classic study

ASSISTANT

Below is a list of Lynn Thorndike discusses (or draws upon) in *A History of Magic and Experimental Science*. These are the famous marquee names (Albertus, Roger Bacon, Ficino, etc.) but rather

that Thorndike used to reconstruct the long prehistory of experimental science through magical, astrological, and technical literature.

I focus on sources that are (a) genuinely mentioned by Thorndike, (b) often overlooked in modern summaries, and (c) represent the

“background radiation” of medieval natural magic, occult experimentation, and technical lore.

Liber Vaccae

A strange Arabic–Latin manual of grotesque magical experiments, including artificial life and homunculus motifs.

Secreta Secretorum technical

Not the moral advice sections, but the embedded alchemical, medical, and physiognomic "experiments" seldom discussed in modern histories.

Transmission of *Cheirokmeta* traditions through excerpted medieval recipe books.

Experimenta

A pseudo-scientific recipe tradition Thorndike

mines for evidence of medieval “experiments.”

Especially lesser Solomonic manuals preserved in Latin miscellanies.

Liber Prestigiorum Heremii

Attributed to Hermes—concerned with illusions, fire tricks, and proto-chemistry.

Astrological–medical conjunction literature passing as Greek authority.

Liber Antimaqui

A rare magical book with instructions for sympathetic operations.

Picatrix

Thorndike used manuscript-only evidence.

Lapidaria

Gem magic treated as natural philosophy.

De Lapidibus

Not the well-known physiognomic corpus but the obscure stone-lore treatise.

Liber de Mirabilibus Mundi

Thorndike treats these “marvels books” as important proto-science.

Virtutes Herbarium

Especially anonymous 11th–12th century compilations.

Liber de Imaginibus

Often preserved only in partial Latin translations.

Circa instans

Used by Thorndike as evidence of technical

precision.

Latin talismanic texts drawing on Hermetic seal theory.

Cyranides

Transmission of animal- and stone-based magical correspondences.

Not the *Liber Introductorius* but the tiny technical fragments.

Arbatel

A surprisingly natural-philosophical “white magic” manual.

Liber Razielis

The angelic cosmography and natural recipe portions.

Hermes de Sigillis

Used by Thorndike to establish continuity with late antique theurgy.

Pseudo-Ramon Lull “experiments” that Thorndike examines critically.

Secretum Philosophorum

A collection of pseudo-experimental tricks, illusions, and recipes.

De rerum natura

Especially zoological marvels that border on magical natural history.

Speculum Naturale

Thorndike draws attention to his inclusion of occult stones, beasts, and wonders.

Liber de Ornatu Mulierum

Cosmetic manuals as proto-chemical texts.

Angelic-scrying magic with astronomical schematics.

De Secretis Mulierum

A medical-magical treatise Thorndike uses to track naturalistic explanations of reproduction.

Lapidary

Not obscure now, but Thorndike foregrounds its magical efficacy sections.

Obscure ritualized instructions embedded in manuscripts.

Triumphal Chariot of Antimony

Thorndike was unusually interested in pre-printed

variant traditions.

Liber ignium ad comburendos hostes

A proto-chemical military manual about incendiaries, once attributed to Marcus Graecus.

Botanical magic with Christian framing.

De mirabilibus mundi

Spurious marvel literature.

De mineralibus

Sections Thorndike considered “proto-experimental.”

De Remediis

Used by Thorndike to show continuity between natural remedies and magical procedure.

He examines ritual instructions preserved in odd codices—not just the Key of Solomon.

Material surviving only in manuscripts Thorndike catalogued.

De secretis naturae

A nexus of natural magic and technical “experiments.”

Liber de rebus mirabilibus audiendi

A little-known marvels book Thorndike treats as evidence for transmission of “experimental” curiosity.

- , with Thorndike's interpretation of its importance
- A showing how these obscure traditions feed into the rise of experimental science
- A used by Thorndike
- A cross-referencing modern scholarship that updates or corrects Thorndike's views

Would you like a or a ?

USER

Summaries ten per response

ASSISTANT

Great — here are the , written in a style that reflects Thorndike's own concerns: transmission, manuscript culture, credulity vs. rationalism, and the continuity he saw between "magic" and the emergence of experimental habits of mind.

(Each entry: , , .)

Liber Vaccae

A bizarre Arabic–Latin collection of magical and pseudo-chemical procedures attributed to Pliny or “the Wise Men of India.” It contains recipes for artificial generation of life, transformations of animals, unguents that induce visions or madness, and processes resembling alchemy. Thorndike treats it not as nonsense but as evidence of medieval belief in , showing how magical handbooks encouraged readers to treat the natural world as malleable and subject to hidden laws. Its grotesqueness is less important to Thorndike than the mindset it cultivates:

Secreta Secretorum

Beyond its famous didactic advice to rulers, this text includes

. Thorndike emphasizes that these practical portions circulated widely among Latin readers and may have been taken quite seriously by early experimenters. The text represents an early instance of

, blending astrology, chemistry, and political science. Thorndike sees it as a bridge from learned magical lore to more systematic observations in medicine and alchemy.

Cheirokmeta

Although known today mostly through Greek fragments, the Latin Middle Ages preserved snippets of material attributed to “Democritus the Magician.” These included recipes for dyes, hardening metals, manipulating precious stones, and performing feats of illusion. Thorndike uses this transmission to argue that medieval readers inherited a
rooted in antiquity. Bolos’s willingness to investigate natural causes aligns strongly with Thorndike’s thesis that “experimental science” emerged from artisanship and magical operations rather than abstract philosophy.

Experimenta

A spurious but influential set of magical and medical recipes attributed to the Aristotelian commentator. These texts offer procedures for healing, enhancing memory, influencing emotions, and producing marvels. Thorndike seizes on this tradition to show how

. Medieval scholars often believed that Aristotle and his commentators had performed practical trials, blurring the line between philosophical credibility and magical experimentation. This created an intellectual climate willing to test recipes or trust others who claimed to.

Solomon

Aside from the later *Clavicula Salomonis*, the Middle Ages preserved a family of obscure Solomonic manuals concerned with

. Thorndike notes that these manuscripts, often in marginal miscellanies, represent a long-lived tradition of

: step-by-step instructions governed by precise natural correspondences. For Thorndike, their importance lies not in demonology but in how they model

, precision, and repeatable protocol — ingredients of experimental practice.

Liber Prestigiorum Heremii

This rare text attributes to Hermes a series of "prestiges," meaning both and (e.g., fire tricks, chemical reactions, mechanical devices). Thorndike highlights it because it blurs magic and mechanical ingenuity, showing that medieval readers did not sharply divide natural magic from early engineering. The text exemplifies the medieval understanding that , a conceptual bridge from occult lore to the mechanical arts later central to Renaissance experimental natural philosophy.

Attributed to the ancient physician Thessalus of Tralles, these works describe how planetary configurations influence the efficacy of medical treatments. Thorndike treats this as an important chapter in the history of , a theory that demanded attention to . Even though based in magical cosmology, the structure of Thessalian recipes encouraged medieval practitioners to attend to natural regularities. Thorndike argues that this sustained a proto-scientific habit of correlating natural conditions with outcomes.

Liber Antimaqui

An obscure magical compilation dealing with sympathetic and apotropaic operations. Thorndike examines it as an example of

, showing how medieval scribes often combined narrative elements with practical instructions. Its significance lies in demonstrating how magical handbooks circulated widely among clerics and were not isolated “occultist” artifacts. For Thorndike, every such text constituted a node in a broader intellectual network that treated and as legitimately investigable.

Picatrix

Thorndike combed through Latin manuscripts of the *Picatrix* to highlight sections never printed or translated in his day, especially the intricate . These tables systematize correspondences between planets, stones, plants, and metals with extraordinary rigor. Thorndike sees this as an early attempt to

. Even if modern readers classify this as superstition, Thorndike argues that the mindset is classificatory and quasi-scientific, showing an impulse toward comprehensive natural explanation.

Lapidary

A lapidary promising miraculous effects from stones — protection, healing, influence, fortune. Thorndike uses texts like this to show that medieval Europe treated minerals as , encouraging empirical manipulation of stones (grinding, heating, engraving). Rather than dismissing this as credulous, Thorndike emphasizes that it shaped early chemical curiosity and craft experimentation. The lapidaries taught people to , anticipate properties, and catalog effects — a proto-mineralogy under the sign of magic.

If you'd like, I can continue with in the next message.

USER

Continue

ASSISTANT

Here are the ,
continuing in the Thorndike-inflected style, with emphasis on manuscript culture, proto-experimental habits, and the continuity between magic and natural philosophy.

De Lapidibus

A short but influential treatise on the occult properties of stones, falsely attributed to Aristotle. Thorndike notes that medieval scholars often read this alongside the more “canonical” Aristotelian corpus, giving it undue prestige. Because it blends mineral lore with and practical instructions, Thorndike treats it as an early model of .

Readers attempted to verify or refine these claims, thus engaging in rudimentary mineral experimentation.

Liber de Mirabilibus Mundi

One of many “Books of Marvels” circulating under Albert’s name, usually including exotic natural wonders, magical stones, strange animals, and recipes for producing marvels. Thorndike treats these not as fantasies but as , often arranged in quasi-scientific categories. Even the more unlikely entries show a desire to understand unusual phenomena. The misattribution to Albertus heightened the book’s credibility and encouraged scholastic readers to integrate magical marvels into natural philosophy.

Virtutes Herbarium

These early medieval compilations list plants with brief but potent claims about their hidden powers — curing diseases, warding spirits, influencing emotions, or inducing dreams. For Thorndike, such herbals are important because they represent

passed among monastic, medical, and artisanal circles. They encouraged empirical procedures: gathering herbs at certain times, preparing concoctions with precise steps, testing effects. Thorndike views them as an early form of

, albeit framed in magical categories.

Liber Imaginum

A significant but under-read work in which astrological images — engraved on metals or stones at specific planetary hours — are said to exert influence on weather, health, and emotions. Thorndike stresses its importance because it shows : rules for making talismans, conditions for success, materials required. Such texts modeled , a feature Thorndike sees as anticipatory of experimental method. They also reveal how Arabic astral magic infiltrated Western natural philosophy.

Though it later became influential, Thorndike emphasizes its lesser-known aspects: pragmatic instructions for identifying, storing, and preparing materia medica. This is not “high magic,” but it inherits the same belief in , now framed in medical terms. Thorndike treats it as evidence that , and that empirically oriented pharmacy grew directly out of traditions that assumed nature’s hidden virtues were manipulable.

Not a single text, but a body of manuscript traditions concerned with believed to channel cosmic forces. Thorndike highlights these because they required precision in engraving, timing, and material selection. Their logic — that nature responds consistently to the right combination of matter and moment — reflects a . Such literature also shows the absorption of late antique Hermeticism into medieval technical practice.

Cyranides

A compendium of correspondences among stones, animals, plants, and letters of the alphabet. Western readers encountered it in partial, often corrupt Latin versions. Thorndike uses this text to illustrate the continuity of from antiquity to the Middle Ages. It is important for him because it presents nature as an interlocking system of sympathies: observing one property helps infer another. This holistic but orderly vision of natural powers became a substrate for later scientific classification.

Beyond his major translations, Michael Scot was credited with numerous shorter, technical works on astrological elections, weather prediction, and divinatory experiments. Thorndike often dwells on these obscure fragments, treating them as windows into the of medieval astrologer-magicians. Because they were circulated as practical, use-oriented manuals, they reveal an applied astronomy that blurred into meteorology and protophysics. Thorndike sees this as “science in the making,” indistinguishable from magic yet committed to prediction grounded in celestial order.

Arbatel de magia veterum

A 16th-century text, but Thorndike treats it as representative of an older magical rationalism. Unlike demonic grimoires, the *Arbatel* presents magic as a form of , emphasizing moral purity, disciplined study, and alignment with divine order. Thorndike sees it as the endpoint of a tradition in which magic becomes , illustrating how Renaissance thinkers reinterpreted magical practice as part of a lawful cosmos. It also contributed to the intellectual climate that nurtured early modern experimenters.

Liber Razielis

This massive angelological encyclopedia contains elaborate cosmologies, lists of natural virtues, astrological correspondences, and ritual instructions. Thorndike is particularly interested in the sections that present

rather than overt mysticism. Its encyclopedic nature made it a reference manual for clerical experimenters who wanted to consult angelic intelligences or manipulate natural forces. Thorndike views it as evidence of how

existed in the same intellectual ecosystem.

If you want, I can continue with

next.

USER

Give me twenty takeaways from his argument and then put them in conversation with more recently scholarly developments and controversies

TOOL

UNKNOWN

{"search_query":[{"q":"Lynn Thorndike History of Magic and Experimental Science main theses magic as precursor of experimental science","recency":365},{"q":"historiography of magic and science Thorndike critique","recency":365},{"q":"Lawrence Principe history of alchemy experimental science continuity","recency":365},{"q":"William Newman alchemy and the origins of modern science","recency":365}]}

TOOL

UNKNOWN

{"search_query":[{"q":"David Pingree \"Lynn Thorndike\" magic experimental science critique","recency":730},{"q":"Katharine Park Lorraine Daston Wonders and the Order of Nature magic science","recency":730},{"q":"Richard Kieckhefer magic in the middle ages overview","recency":730},{"q":"Wouter Hanegraaff esotericism historiography science magic","recency":730}]}

ASSISTANT

Here are

, each followed by how later

scholarship has

that move.

I'll do it as for each point.

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same

His central move is that you *cannot* write a serious history of medieval and early modern science without magic, astrology, and alchemy. They're not peripheral superstitions; they're integral to how people investigated nature. cite turn0search0 turn0search28

This has basically *won*. Historians of alchemy and "occult sciences" (Principe, Newman, Pingree, etc.) explicitly position themselves as extending or correcting Thorndike rather than rejecting him. Principe's "Alchemy Restored" and *The Secrets of Alchemy* are explicit projects to reinsert alchemy into mainstream history of science, arguing it was a key experimental discipline. cite turn0search22 turn0search2

He insists medieval magic, astrology, and alchemy follow a — they're not random insanity. Practitioners reasoned from premises inherited from Aristotle, Galen, Scripture, and late antique Platonism.

Kieckhefer's "Specific Rationality of Medieval Magic" explicitly names Thorndike and builds on this idea: magic has its own , distinct but not irrational. cite turn1search17 turn1search5 Hanegraaff's work on esotericism likewise argues that so-called "rejected knowledge" is organized and intellectually serious, shaped by polemics rather than being inherently irrational. cite turn1search32 turn1search36

He argues that “experimental science” did not suddenly appear in the 17th century; its roots lie in , natural magic, and alchemy—people trying things out in the lab, garden, workshop, or pharmacy. cite turn0search0 turn0search24

Newman’s *Atoms and Alchemy* and his broader work on “chymistry” sharpen this: specific alchemical traditions developed *operational* criteria for elements and corpuscles, directly feeding later mechanistic matter theory and Boyle’s chemistry. cite turn0search7 turn0search23 Principe’s reconstructions of alchemical experiments strengthen Thorndike’s intuition with lab evidence: many alchemists really were doing sophisticated experimental work. cite turn0search14 turn0search22

Much of his “argument” is actually a massive : identifying, dating, and describing hundreds of texts (including weird things like *Liber vaccae*). This is an implicit claim: before theorizing, you must know *what* the sources are.
cite turn1search0 turn1search29

Modern work on Liber vaccae, Picatrix, ars notoria, etc., often begins by saying “Thorndike first mapped this terrain.” A good example: Van der Lugt on *Liber vaccae* explicitly credits Thorndike’s pioneering manuscript work.
cite turn1search29
So on this front, he’s still the big scaffolding behind more interpretive work.

He treats astrology not as marginal superstition but as a of the medieval curriculum: built on astronomy, geometry, and complex calculations, and heavily used in medicine and politics. cite turn0search24 turn0search33

Modern historians of astrology (e.g. Pingree and those cited in “Notion of Occult Sciences” discussions) broadly agree that astrology was a technical science in its own terms and a central part of learned culture. cite turn1search33 The controversy now is less *whether* it was central and more about how to balance its intellectual, social, and political functions.

He pushes hard that alchemy involved , not just weird symbolism or psychological allegory. He links it directly to the emergence of experimental methods, especially in the 16th–17th centuries (Boyle, etc.).
cite turn0search24 turn0search17

Principe and Newman have basically turbocharged this, arguing that alchemists were central to the development of chemistry, not a sideshow. They show that alchemists developed corpuscular theories and techniques that Boyle and others directly used.
cite turn0search2 turn0search23
The controversy is now mainly with older, more dismissive views (e.g. Jungian "it's all allegory"), *not* with Thorndike.

He repeatedly shows clerics copying, using, and sometimes defending magical texts (Solomonic, ars notoria, angel magic, etc.). The Church represses *some* things, but is also a major

.

cite turn1search25 turn0search24

Kieckhefer's *Magic in the Middle Ages* and work on Christian ritual magic make this much more nuanced: ecclesiastical ritual and "magic" sit on a continuum; inquisitors, theologians, and ritual magicians share assumptions. cite turn1search5 turn1search9 Hanegraaff likewise shows that "esotericism vs. orthodoxy" is a constructed boundary shaped by confessional polemics. cite turn1search32 turn1search16

He is fascinated by “marvels books” and miracle-like phenomena (strange stones, monstrous births, odd optical effects). He treats wonder as a *driver* of investigation: marvels prompt classification and causal speculation. cite turn0search0 turn0search24

Daston & Park’s *Wonders and the Order of Nature* gives this its full theoretical spin: wonder and “wonders” structure natural inquiry between 1150 and 1750, and the style of wonder changes over time. cite turn1search14 turn1search1 They move beyond Thorndike’s mostly descriptive approach, arguing that the *affect of wonder* is central to how naturalists conceptualized “nature” itself.

He devotes huge space to writers like Vincent of Beauvais, Thomas of Cantimpré, Albertus, and pseudo-Albertine compilations, arguing that encyclopedias and “books of secrets” are crucial sites where magical and scientific lore are

.

cite turn0search24 turn0search28

Recent work on “books of secrets,” household recipe books, and artisanal knowledge (Smith, Eamon, etc.) echoes this but pushes it down into

as well. Daston & Park also treat collections of wonders as epistemic instruments, not just trivia dumps. cite turn1search14 turn1search18

His archive is overwhelmingly

in institutional libraries. He’s explicit that he’s following the “learned tradition,” which means he underplays vernacular, oral, and popular magic.

Kieckhefer, Bailey, and others put much more emphasis on

, showing how elite and popular practices intersect.
cite turn1search5 turn1search31
So Thorndike’s *framework* is still influential, but his archive looks narrow compared to current work that includes archaeology, material culture, and judicial sources.

Despite his sympathy for medieval thinkers, he still has a fairly narrative: over time, the “experimental” side of magic is retained and rationalized; the more superstitious or demonic elements drop away, culminating in “real science.”

The “Scientific Revolution” literature tends to be more suspicious of simple progress narratives. Daston & Park, Shapin, Dear, etc. stress of scientific styles rather than a straight line of progress. cite turn1search1 turn1search26 Principe and Newman accept continuity but downplay any triumphalist story that sees magic purely as something to be transcended. cite turn0search22 turn0search23

He’s classic early-20th-century intellectual history: ideas, texts, and learned figures dominate; printing, patronage, artisanal economies, and state formation play a minimal role.

Historians like Newman (with attention to artisanal practice), Principe (recreating lab work), and the broader “practice turn” in history of science shift focus to

.

cite turn0search14 turn0search23 Historians of magic and medicine similarly integrate legal, economic, and gendered contexts, complicating Thorndike’s text-heavy portrait.

He tends to classify magic alongside astrology and alchemy as , distinct from theology and devotion. Religious frameworks matter, but he treats magic mainly as a form of natural philosophy rather than as religious practice. cite turn1search0 turn0search24

The esotericism field (Faivre, Hanegraaff, etc.) shows that , and that “occult sciences” are themselves a category produced in early modern polemics and Enlightenment historiography. cite turn1search32 turn1search20 Kieckhefer’s work on Christian ritual magic also blurs Thorndike’s neat line between theology and occult practice. cite turn1search9 turn1search5

He constantly notes how medieval thinkers assume —hidden virtues in stones, herbs, metals—that produce effects not obvious from surface properties. For him, this is the conceptual core linking magic to natural philosophy. cite turn0search0 turn0search28

New work on “occult qualities” (in both scholastic philosophy and chymistry) often shows that these are , not lazy gap-fillers: they’re linked to debates on causality and matter theory. Newman’s work, for example, shows alchemists turning occult behavior into micro-mechanistic explanation. cite turn0search23 turn0search15

He treats medical astrology, image magic, and talismans as *major* fields where celestial theory and practical health care meet. They show how far “scientific” medicine is intertwined with astral lore. cite turn0search24 turn1search21

Historians of medicine and astrology generally confirm this integration and deepen it: studies of astral medicine, hospitals, and learned physicians stress that astrological diagnosis and timing were routine, not fringe. The present debate is more about than about whether Thorndike was right to see this as central. cite turn1search33 turn1search5

He reads early modern figures like Boyle, Kepler, and Newton as : still soaked in astrology, alchemy, and biblical chronology while also pioneering “modern” methods. Science grows out of magic rather than abruptly replacing it.
cite turn0search17 turn0search29

This idea has been enthusiastically taken up and sharpened—especially for Boyle and Newton. Principe’s *Aspiring Adept* argues that Boyle was actively alchemical; Newman shows how chymistry underpins his mechanical philosophy.
cite turn0search10 turn0search23
For Newton, studies of his alchemy and theology likewise echo Thorndike’s intuition that the “founders” of modern science are deeply entangled with magical and theological projects.

He is quite about Hermetism, Rosicrucianism, and similar trends. He doesn't deny their importance, but he resists reading them as bearers of timeless mystical wisdom. For him they're historical, often confused, sometimes credulous.

Post-Yates debates about Hermeticism and the "Yates thesis" (Renaissance Hermetism as root of modern science) often use Thorndike as an earlier, more skeptical foil. Esotericism scholars today try to steer between Yates-style over-celebration and Thorndike's cooler stance, emphasizing both the of Hermetic/esoteric currents and the dangers of mythologizing them.
cite turn1search16 turn1search32

By spending energy on *Liber vaccae*, obscure Solomonic manuals, talismanic tables, cosmetic books, and secret books, he implicitly argues that

, not just curiosities.
cite turn1search29 turn1search25

This has become standard; scholars now write whole monographs on things like *Picatrix*, *Liber vaccae*, ars notoria, or individual grimoires. Thorndike's sense that the "weird little stuff" reveals the is very much in line with current micro-historical and material-culture approaches.
cite turn1search25 turn1search21

Although he doesn’t use the language, his practice shows magic as a where medicine, astronomy, theology, philosophy, and artisanal crafts meet.

Kieckhefer literally describes magic as a “crossroads” of medieval culture, making explicit what Thorndike does implicitly. cite turn1search23 turn1search35 Daston & Park similarly treat marvels as points where natural philosophy, theology, and courtly culture intersect. cite turn1search14 turn1search18

His eight volumes are essentially a

: map the terrain of texts, show that magic belongs in the history of science, hint at continuities, and leave reinterpretation to the future.

Modern debates about:

- the “Scientific Revolution” vs. long continuities,
- the status of alchemy and astrology,
- how to define “esotericism,”
- how to integrate social/gender/material histories—

all play out on a map Thorndike helped draw. Scholars now critique his progressivism, his narrow social lens, and his focus on Latin elites,

but almost nobody goes back to the old position that magic is irrelevant to the history of science. On that core point, Thorndike “wins,” and the controversies are over to tell that entangled story, not *whether* to tell it.
cite turn1search17 turn1search9

If you’d like, next step could be:

- Turning this into an (Thorndike → Yates → “practice turn” → esotericism studies), or
- Zooming in on one axis, like or , and sketching arguments you could use in a paper.

USER

Give me 40 topics in recent history of science scholarship on magic for grad students to use for finding projects

ASSISTANT

Here are in the *recent* history of science scholarship on magic, astrology, and alchemy.
Each is crafted to be but .

These reflect developments from roughly the last 20–30 years of scholarship across history of science, religious studies, and esotericism studies.

I've grouped them thematically to help you browse.

Inspired by Principe & Newman: how replication impacts interpretation of historical texts.

Continuities between workshops, guild culture, and laboratory technique.

How alchemists used diagrams, emblems, and color stages to encode procedure (Kassel, Forshaw).

How “secrets” structured communities of trust and exclusion.

Paracelsians, medical alchemy, and the institutionalization of chemical cures.

Studying alembics, athanors, and tools through archaeology or museum collections.

Mining, metal transmutation, and the blurred boundary between craft and esoteric theory.

Dyestuffs, pigments, and proto-industrial experimentation.

Looking at shared labor, gendered work, apprentices, and the family-lab structure.

New interpretive work combining alchemical exegesis with mathematical praxis.

Curriculum, mathematics, instrument use, and student practice.

How astrological timing shaped public health strategies.

New work on Picatrix, Ikhwan al-Safa', and Thābit ibn Qurra in Arabic manuscripts.

Their role in diplomacy, weather prediction, and crisis management.

Using manuscript stemmata to track doctrinal shifts.

Astrolabes, celestial globes, and the magical uses of scientific instruments.

How scholars debated, defended, or critiqued astrology amid confessional tension.

Connections between omen interpretation and proto-statistical thinking.

Ars notoria, angelic communication, and cathedral-school pedagogy.

How inquisitors and theologians defined the boundaries of licit/illicit knowledge.

Paper, ink, diagrams, binding, and the ontology of “charged” objects.

Demonic causation theories and the problem of explaining invisible agents.

Post-Yates scholarship focusing on Pico, Reuchlin, and Renaissance Hebraism.

Charm books, amulets, and pilgrimage cures as “practical theology.”

Monks as compilers, experimenters, and keepers of forbidden knowledge.

Prayer experiments, visionary training, and spiritual technologies.

How hidden virtues, forms, and species shaped theories of causality.

Descartes, Leibniz, Spinoza, and their selective incorporation of occult sciences.

How magical phenomena forced scholars to rethink what counts as lawful nature.

Recipe books, household science, and the democratization of experiment.

How philological scholarship transformed the

interpretation of Hermetica, Pseudo-Aristotle, and alchemical texts.

How secrecy, initiation, and hidden knowledge intersect with state power or censorship.

Cultural, mathematical, and institutional explanations.

Reassessing Weber through the lens of magical survivals and scientific magic.

Comparative history of experiment across civilizations.

Cross-cultural transmission of horoscopy and mathematical astronomy.

Situating divination and healing traditions within global history of science.

Magic, medicine, and negotiation of knowledge authority under empire.

How different cultures categorized the strange and the marvelous.

Hermetic, Daoist, and Tantric knowledge systems as alternative scientific epistemologies.

I can turn this into:

- A for a graduate seminar
- A of key books and articles for each topic
- A set of tailored to your own interest in alchemy, esotericism, pedagogy, or digital humanities
- A flowchart guiding students to topics based on languages, methods, and archival access.

Would you like that?

USER

Instrument design

ASSISTANT

Here’s a focused, graduate-level research guide on

—one of the richest, fastest-growing subfields where students can find publishable projects. I’ll give you:

1.
2. (astrological, alchemical, divinatory, ritual)
3.
4.

This is tuned to the current cross-field convergence of history of science, material culture studies, religious studies, and the anthropology of technology.

Most modern treatments focus on scientific navigation and measurement. But medieval and Islamic astrolabes were also *ritual and talismanic devices*.

Analyze astrolabe plates engraved for horary astrology, talismanic elections, or image magic.

Picatrix

The *Picatrix* contains instructions for constructing engraved images, mirrors, and metal objects tuned to planets.

Reconstruct one talismanic instrument using period metallurgy and evaluate the interplay between magical and technical constraints.

Recent scholarship shows alchemical ovens, retorts, and alembics were highly engineered.

Study how furnace design changed experimental reproducibility in early modern chymistry.

Mirrors were used not just for optics but for necromancy, angelic contact, and astral projection.

Compare optical treatises (e.g. Alhazen) with ritual manuals for scrying mirrors.

Magical texts often include mandalas, concentric circles, geometric talismans, and grimoire diagrams. These diagrams *functioned* like instruments for mental visualization and ritual precision.

Analyze how geometric design embeds cosmology and operational rules.

Many survive in Africa, the Islamic world, and medieval Europe.

Study how material constraints (size, portability, marks, layout) shaped epistemic authority.

Ars Notoria

Recent work treats Dee's equipment (shewstone, table of practice, sigillum Dei Aemeth) as a *theological–technical system*.

Examine how ritual instruments were designed to stabilize visionary experience.

Surgical and magical instruments were often co-designed: lancets engraved with zodiacal symbols, phlebotomy charts aligned with lunar phases.

Reconstruct the astrological logic built into medical tool design.

Islamic astral magic describes highly detailed fabrication of lamps, statues, talismanic machines, and moving automata.

Study engineering principles embedded in magical automata design.

Kircher, Fludd, and others designed instruments that embodied Hermetic cosmology—hydraulic organs, catoptric theaters, magnetic displays.

Analyze how magical cosmologies were encoded into mechanical design.

- Astrolabes (scientific, talismanic, and judicial)
- Quadrants and horary devices
- Celestial globes with zodiacal projections
- Geomantic tablets and sand tables
- Scrying mirrors (obsidian, polished metals, black glass)

- Athanors (long-burn furnaces)
- Alembics (distillation heads)
- Retorts, cucurbits, pelicans
- Sublimation towers
- Laboratory balances and crucibles
- Color-testing instruments (alchemical “tables of colors”)

- Enchanted circles and drawn geometries (as spatial instruments)
- Sigils, pentacles, lamens
- John Dee’s table of practice and wax seals
- Divination rods, lots, and casting tools
- Angelic or talismanic discs engraved according to planetary hours

- Catoptric “demon mirrors”
- Crystal or glass spheres for visionary practice
- Kircher’s speaking tubes and acoustic devices
- Magical automata (Islamic, Byzantine, Renaissance)
- Rotating cosmograms used for trance induction

These topics sit at the intersection of several influential currents in current scholarship:

Historians now treat instruments not as passive tools but as .
→ For magical instruments, this means treating a talisman or furnace like a *knowledge-generating* device, not superstition.

Instead of focusing only on abstract theories, scholars examine what practitioners *did* with their hands.
→ Alchemical ovens, scrying mirrors, or astrological boards become evidence of .

Talismanic instruments in Islamic, Jewish, African, and Asian traditions contribute to a narrative of science, not a Eurocentric one.

Work by Hanegraaff, Forshaw, and others treats magical technologies as part of intellectual history, not marginal curiosities.

Mirrors, diagrams, ritual spaces, and sensory manipulation techniques are being reinterpreted as for producing knowledge or visions.

The line between magical automata and early engineering is being redrawn (e.g., Islamic automata and Renaissance theatrical machinery).

Here are some scaffolds that could turn into thesis chapters:

- What counts as “measurement” in a magical context?
- How do magical instruments mediate between visible and invisible causal layers?
- Can diagrams and ritual geometries be treated as “virtual instruments”?
- How did artisans translate astrological or alchemical theory into metal, glass, or wood?

- How did instrument design reflect the availability of local materials or guild expertise?

- What happens when an instrument fails? What does that failure reveal about theory and practice?

- How do Islamic talismanic devices compare with European ones in terms of mathematical rigor?

- Are there global parallels between scrying devices, spirit-calling technologies, or visionary media?

- How did instruments authorize a practitioner's status as a "wise man," "philosopher," or

“magus”?

- How do magical instruments function within patronage systems, courts, or monastic settings?

- How did sensory experience (light patterns, reflection, heat, color) help induce altered states?

- How does instrument use structure the user’s embodied engagement with cosmology?

I can produce:

- on magical instrument design
- for a semester-long graduate seminar on instruments in magic and early science
- for each instrument type
- for students choosing thesis topics

Just tell me which direction you want to go.

USER

Give me a timeline with moments from Thorndike and Five topics per century

ASSISTANT

Below is a highlighting:

1. — key authors, texts, discoveries, or intellectual shifts that Thorndike foregrounds in *A History of Magic & Experimental Science.*
2. — drawn from (Principe, Newman, Kieckhefer, Daston & Park, Hanegraaff, Pingree, etc.) to help grad students find projects framed around that era.

This gives you a from antiquity through the early modern period.

- Bolos of Mendes (Pseudo-Democritus): early alchemical recipes, magical naturalism.
- Hermetic corpus and technical Hermetica (astrology, talismans, cosmology).
- Hellenistic astrology (Claudius Ptolemy, Dorotheus of Sidon).
- Late antique medical magic: Thessalus, Galen's occult materia medica.
- *Cyranides* tradition: correspondences of stones, animals, plants.

1. [illegible] into Arabic science (manuscript stemmata).
2. [illegible] (furnaces, dyes, crucibles).
3. [illegible] and the origins of astral “science.”
4. [illegible] use of amulets, phylacteries, sympathetic cures in Graeco-Roman hospitals.
5. [illegible] Hermetic cosmology vs. early Gnostic scientific epistemologies.

- Byzantine compilations of wonders and marvels.
- Early Christian debates on demons, angelic causation, and magical practice.
- Late antique lapidaries and herbal magical texts.
- Emergence of ritual magic traditions embedded in Christian monastic life.
- Preservation of technical alchemy through Byzantine redaction.

1. (saint cults, relics, exorcistic technologies).
2. and continuity with Hellenistic metallurgical practice.
3. (ars notoria, healing charms).
4. as proto-scientific encyclopedias.
5. through Syriac scholars (e.g., medical astrology in monastic hospitals).

- Arabic translations of Greek technical sciences (alchemy, astrology, medicine).
- Thābit ibn Qurra: mathematical astronomy + talismanic image theory.
- Early Islamic alchemical synthesis (Jābir ibn Hayyān).
- Transmission of Hermetic ideas into Arabic philosophy.
- Carolingian interests in computus and astral timing.

1.
astrolabes, magical mirrors, horoscopes as tools.
2. as a mathematical technology (Picatrix precursors).
3. in early Islamic image magic.
4. in Jābirian sulfur-mercury theory.
5. computus, prognostication, and agrarian astronomy.

- Latin Europe receives fragments of Greco-Arabic occult knowledge through Spain and Italy.
- Re-emergence of natural magic in monastic settings (herbals, charms, medical recipes).
- Development of early medieval bestiaries and stone-books.
- Manuscript diffusion of *Liber Vaccae* and other grotesque experiment-books.
- Growth of cathedral-school interest in natural wonders.

1. [illegible] as proto-experimental cultures.
2. [illegible] before the 12th-century translation movement.
3. [illegible] in “minor” magical miscellanies.
4. [illegible] in structuring natural knowledge.
5. [illegible] herbalism, charms, bloodletting manuscripts.

- Massive translation of Arabic science → Latin.
- *Picatrix* enters the Latin tradition.
- Adelard of Bath introduces Islamic science and astrology.
- Development of scholastic views on occult causation.
- Formation of new encyclopedic genres (e.g., *De rerum naturis*).

1.
(astrolabes, quadrants) in cathedral schools.

2. ***Picatrix*** : talismans, elections, ritual engineering.
3. : institutional control of occult knowledge.
4. as data visualization tools.
5. how translators shaped magical epistemology (e.g., Gerard of Cremona).

- Scholastic debates about occult qualities, forms, celestial influence.
- Albertus Magnus synthesizes natural philosophy with magical naturalism.
- Roger Bacon’s experimental program tied to optics and image magic.
- Judicial astrology becomes highly systematized.
- Rise of the *Ars Notoria* and angelic ritual technologies.

1. (mirrors,

lenses, Baconian “visual rays”).

2. (Paris, Oxford).

3. ***Ars Notoria*** for accelerated learning.

4. : horary boards, volumetric astrolabes.

5. : lunar calendars, phlebotomy instruments, astrological surgery.

- Astrology reaches peak institutional power in courts and universities.
- Refinement of alchemical techniques; emergence of vernacular alchemy.
- Proliferation of talismanic and geomantic manuscripts.
- Intellectual tensions between determinism and free will.
- Early critiques of magic within theology, but heavy usage persists.

1.

(e.g., astrology in Florence or Avignon).

2. and household experimentation.

3. and divinatory boards.

4. : astrological-medical attempts to model contagion.

5. (midwifery, charms, domestic tools).

- Renaissance humanism reshapes magical texts (critical editions, new attributions).
- Ficino revives Hermetic and astral magic with philosophical legitimacy.
- Printing spreads “books of secrets.”
- Flourishing of image magic, talismanic theory, and astral medicine.
- Magical engineering (automata, perspective machines) grows in Italy.

1. (Ficino, Pico, pseudo-Aristotle).

2. as vernacular technical literature.
3. blending magic, optics, and engineering.
4. in courts (Sforza, Medici).
5. in ritual and talismanic technologies.

- Paracelsian revolution in medical alchemy.
- Dee and Kelley: angelic technologies and instruments.
- Proliferation of occult philosophy (Agrippa, Bruno).
- Natural magic reframed as scientific curiosity.
- Witchcraft discourse emerges alongside learned magic.

1. (shewstone, table, sigillum) as epistemic machines.
2. and chemical

therapies.

3.

4.

(Della Porta, Bruno).

5.

- Boyle’s engagement with alchemy and experimental method.
- Newton’s massive alchemical corpus.
- Decline of judicial astrology but persistence of astral reasoning.
- Mechanical philosophy inherits occult qualities under new names.
- Magic becomes reframed rather than eliminated.

1. : corpuscular

alchemy, prisms, prophecy.

2. and alchemical failures.

3. and early meteorology.

4.

5. how experimental philosophers reframed marvels.

(Thorndike’s story ends earlier, but scholars continue the line.)

1. in Enlightenment chemistry.
2.

3. in Rosicrucian and masonic circles.
4. (crystal gazing, magnetic mirrors).
5. reshaping European concepts of science and magic.

I can turn this into:

- A ,
- A ,
- A ,
- Or a connecting Thorndike to modern historiography.

Just tell me the format you want.